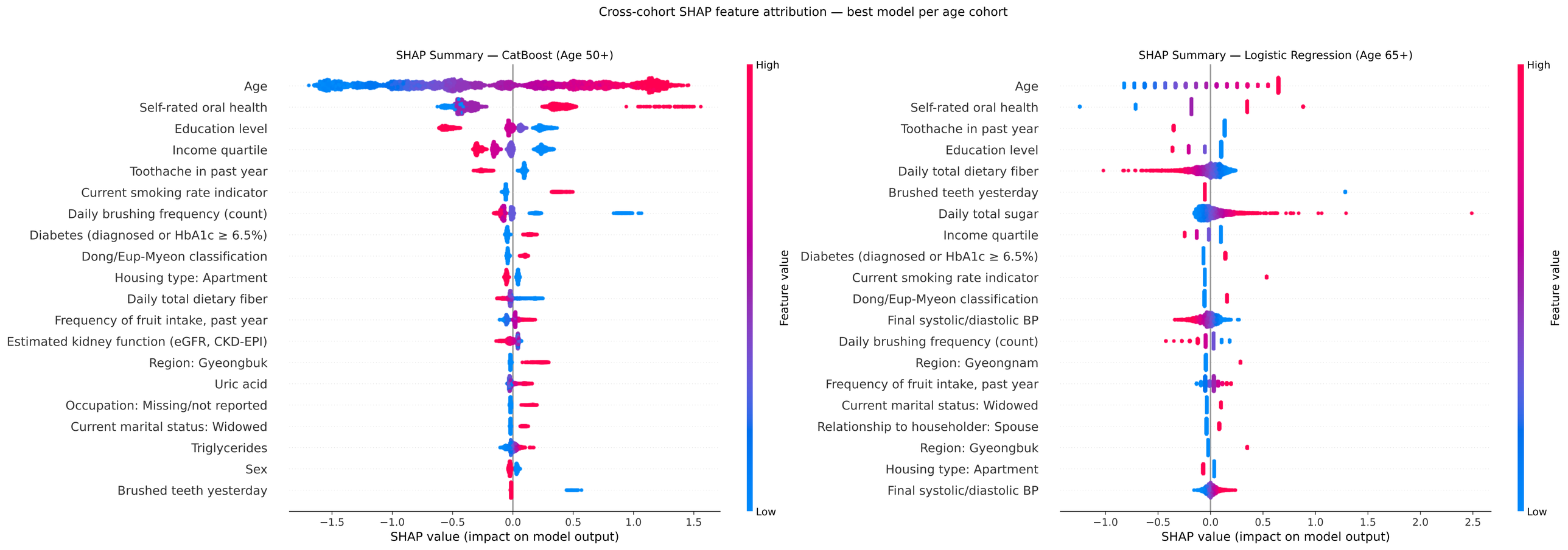

Cross-cohort SHAP feature attribution — best model per age cohort
SHAP Summary — CatBoost (Age 50+)
Age
Self-rated oral health
Education level
Income quartile
Toothache in past year
Current smoking rate indicator
Daily brushing frequency (count)
Diabetes (diagnosed or HbA1c ≥ 6.5%)
Dong/Eup-Myeon classification
Housing type: Apartment
Daily total dietary fiber
Frequency of fruit intake, past year
Estimated kidney function (eGFR, CKD-EPI)
Region: Gyeongbuk
Uric acid
Occupation: Missing/not reported
Current marital status: Widowed
Triglycerides
Sex
Brushed teeth yesterday
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
SHAP value (impact on model output)
High
Feature value
Low
SHAP Summary — Logistic Regression (Age 65+)
Age
Self-rated oral health
Toothache in past year
Education level
Daily total dietary fiber
Brushed teeth yesterday
Daily total sugar
Income quartile
Diabetes (diagnosed or HbA1c ≥ 6.5%)
Current smoking rate indicator
Dong/Eup-Myeon classification
Final systolic/diastolic BP
Daily brushing frequency (count)
Region: Gyeongnam
Frequency of fruit intake, past year
Current marital status: Widowed
Relationship to householder: Spouse
Region: Gyeongbuk
Housing type: Apartment
Final systolic/diastolic BP
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
SHAP value (impact on model output)
High
Feature value
Low